IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2016

EEE/EIE PART III/IV: MEng, BEng and ACGI

Corrected copy

DIGITAL SIGNAL PROCESSING

Wednesday, 14 December 9:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : W. Dai

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Questions

DIGITAL SIGNAL PROCESSING

1. a) Sketch the block diagram of a 2-band maximally decimated filter bank structure showing the analysis and synthesis filter banks connected directly in cascade. [3]

Discuss the effects of aliasing in a QMF maximally decimated 2-band filter bank. Include an explanation for aliasing in terms of the aliasing component matrix $\mathbf{H}(z)$. Include supporting analysis. Include a comparison to the case of oversampled filter banks. [5]

Derive the aliasing cancelling conditions. [2]

b) Consider a linear time-invariant system output signal $y(n)$, input signal $x(n)$ and impulse response $\{h_k\}$ for $k = 0, \ldots, N$ given by

k	h_k
0	-0.0087
1	0.0000
2	0.2518
3	0.5138
4	0.2518
5	0.0000
6	-0.0087

and

$$y(n) = \sum_{k=0}^{N} h_k x(n-k).$$

The frequency response of this system is shown in Fig. 1.1.

Use this system to design aliasing cancelling 2-band maximally decimated QMF analysis and synthesis filter banks. Your design should include the coefficients of all filters. [7]

Describe fully the reconstruction properties of this filter bank and state the group delay of the analysis and synthesis filter banks. [3]

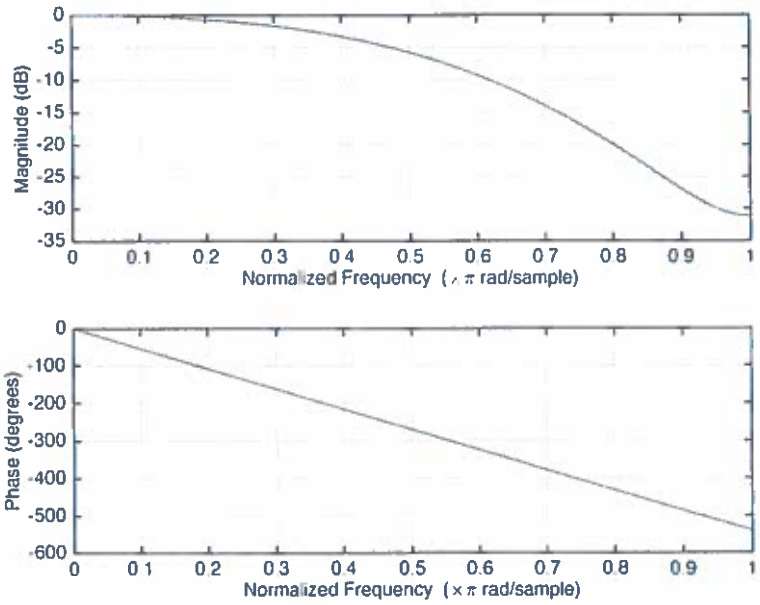


Figure 1.1 Frequency Response

2. a) Show that a 4-point DFT can be expressed in terms of two 2-point DFTs using the Decimation-in-Time FFT approach and write down the recombination equations. [6]

Illustrate your solution using a signal flow graph. [4]

b) Determine the percentage reduction in the number of real multiply operations when a 1024-point DFT is computed using the Decimation-in-Time FFT algorithm instead of a direct implementation of the DFT. [4]

c) Find the magnitude and phase spectra of the discrete-time signal

$$p(n) = [\ 0.1 \quad -0.1 \quad 0.0 \quad -0.1\ ].$$

[6]

3. Consider a discrete-time sequence $x(n)$ having z-transform $X(z)$.

a) State the expression for $X(z)$ in terms of $x(n)$ and explain what is meant by the *Region of Convergence* in this context. [4]

b) Describe the similarities and differences between the z-transform and the Laplace transform. Illustrate your answer using relevant sketches. Comment on the way in which the spectrum of discrete-time signals is represented in the z-domain. [5]

c) i) A particular linear time-invariant discrete-time system is described by the difference equation

$$y(n) = 0.2y(n-1) + 2x(n).$$

Using z-transform relationships, find expressions for the system function $H(z)$ of this system, and the unit impulse response $h(n)$ of this system. [4]

ii) Consider the signal

$$x(n) = \begin{cases} a^n, & 0 \le n \le M-1 \\ 0 & \text{otherwise} \end{cases}$$

where M is an integer and $0 < a < 1$.

Determine the z-transform $X(z)$ of the signal $x(n)$ for any integer value of M.

Sketch in the z-plane a representation of $X(z)$ for the particular case of $M = 8$. [4]

iii) Find the inverse z-transform of

$$G(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$$

for the 3 cases of the ROC:

$$|z| > 1,$$
$$|z| < 0.5,$$
$$0.5 < |z| < 1.$$

[3]

4. A single stage of a lattice structure is shown in Fig. 4.1.

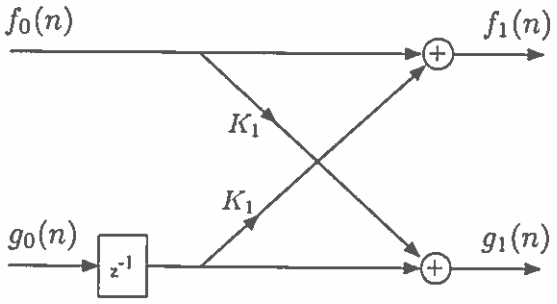


Figure 4.1 Single Stage Lattice Structure

a) Write expressions for $f_1(n)$ and $g_1(n)$ in terms of $f_0(n)$ and $g_0(n)$. [2]

b) Defining $x(n)$ as an input signal that is connected to both $f_0(n)$ and $g_0(n)$, and taking the output $y(n)$ as the signal $f_1(n)$, show that the structure implements a first order FIR filter and write down the difference equation for the filter in terms of $x(n)$, $y(n)$ and the filter coefficients denoted a_k, for integer k. Include formulae for a_k. [4]

c) Sketch a block diagram of two single stage lattice structures in cascade. The first stage is as given in Fig. 4.1 with $x(n)$ as an input signal that is connected to both $f_0(n)$ and $g_0(n)$, as in part b). The second stage uses K_2 in place of K_1 in the first stage. Denote the f and g outputs of the second stage $f_2(n)$ and $g_2(n)$ respectively.

Give expressions for $f_2(n)$ and $g_2(n)$ in terms of $f_1(n)$ and $g_1(n)$. [3]

d) Hence show that the cascaded structure implements a second order FIR filter and write down the difference equation for the filter in terms of $x(n)$, $y(n)$, K_1 and K_2.

Give expressions for the corresponding filter coefficients a_1 and a_2 in terms of K_1 and K_2. [5]

e) Now consider a cascade of m single stage lattice filters for any integer m. Denote this filter's transfer function

$$A_m(z) = \frac{Y(z)}{X(z)} = \frac{F_m(z)}{F_0(z)}.$$

The filter with coefficients b_k and transfer function $B_m(z)$ is defined as

$$B_m(z) = \frac{G_m(z)}{X(z)}.$$

Write down a general expression for b_k in terms of a_k and determine $B_m(z)$ in terms of $A_m(z)$. [6]

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